

Loops (for, while, do-while)

DEF: Loops let you repeat a block of code multiple times without rewriting it.

How to use: `for(int i=1; i<=5; i++){`
`S.O.P(i); }` // for loop

`int i=1;`
`while(i<=5){`
`S.O.P(i);`
`i++; }` // while loop

`int j=1;`
`do{`
`S.O.P(j);`
`j++;`
`} while(j<=5);` // do-while loop (runs at least once)

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When to use :

- for → when you know how many times to repeat.
- while → when you repeat until a condition becomes false (unknown count).
- do-while → when the code must run at least once, regardless of condition.

where to use : printing patterns, processing arrays, repeating calculations, reading multiple inputs.

Why three types instead of one : each fits a dif scenario. Using for when you don't know the count is awkward, using while when you must guarantee at least one run doesn't work - do-while solves that specific case.

Real-life ex : for is like doing 10 pushups (fixed count). while is like eating until you're full (unknown count, condition-based). do-while is like tasting a dish at least once before deciding if you want more.